

Lab 4 Data Flow Diagrams and Decomposition

Supporting tools: Visual Paradigm | Duration: 2 hours and 30 minutes

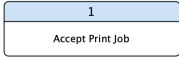
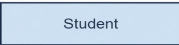
Lab objectives

- Develop a Level 1 DFD from the supplied APS context diagram.
- Identify the processes and data stores needed.
- Decompose the auction process into a balanced Level 2 DFD.

Course learning outcome (CLO 2): Elicit, analyze, prioritize and document requirements for a software system.

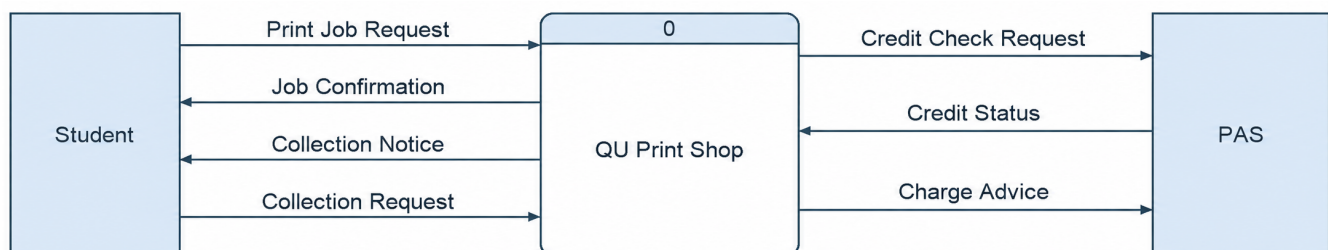
Part A QU Print Shop

Work through A.1 to A.5 with the instructor (25 minutes). Use this worked example as a guide for APS in Part B. Bring the APS case and your Lab 3 work.

Element	Gane-Sarson symbol	Naming example
Process		1 Accept Print Job
External entity		Student or PAS
Data store		D1 Print Jobs
Data flow		Print Job Request

Use these rules throughout the lab: each process needs inputs and outputs; every flow must connect to a process. Label flows with data names. Show store reads and writes separately. Process IDs show hierarchy, not execution order.

A.1 Review the supplied context diagram



The system has two external entities and seven boundary flows. PAS owns credit balances and handles payments. The context is Level 0; its process ID is 0. Internal processes and stores appear in Level 1.

A.2 Identify processes and stores

ID and process	Responsibility
1 Accept Print Job	Check credit and record accepted jobs.
2 Fulfil Print Job	Print jobs, notify students and advise PAS of charges.
3 Record Collection	Record when a student collects a printed job.

Store	Contents	Read by	Written by
D1 Print Jobs	Job ID, document, student ID, printing options and status	2 and 3	1, 2 and 3

A.2 Internal data flows

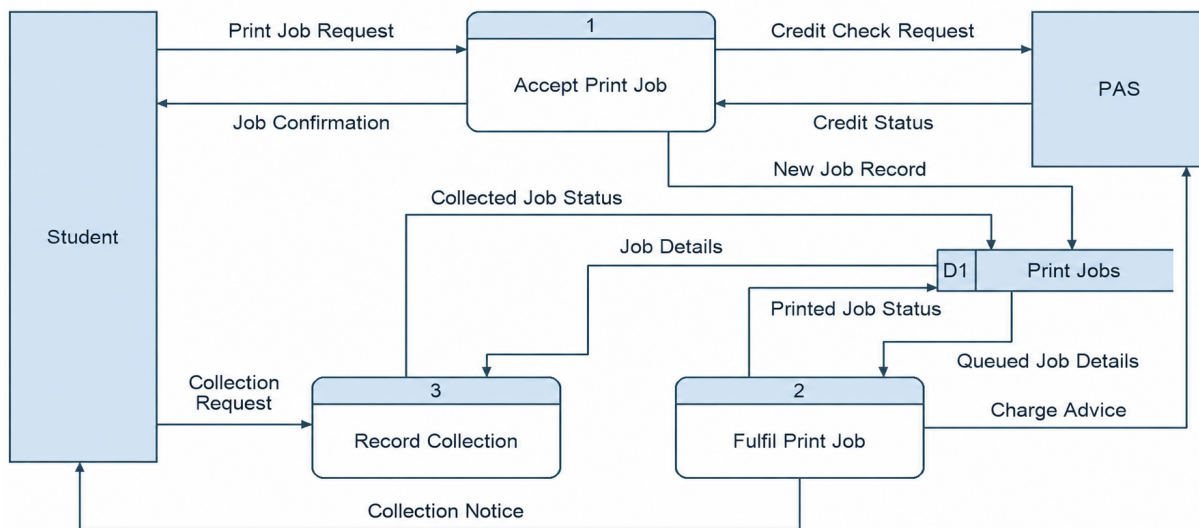
Source	Data flow	Destination	Contents
1	New Job Record	D1	Job ID, document, student ID, options, queued status
D1	Queued Job Details	2	Job ID, document, student ID, options, queued status
2	Printed Job Status	D1	Job ID and printed status
D1	Job Details	3	Job ID, student ID and current status
3	Collected Job Status	D1	Job ID and collected status

At Level 1, the processes share data through D1. Each table row becomes a labelled arrow in the diagram. At Level 2, subprocesses also pass data directly to one another, as shown below.

Example assumption: the Print Shop calculates charges from the document, options and fixed printing rates. Credit Check Request carries student ID and estimated charge; Charge Advice carries student ID, job ID and final charge.

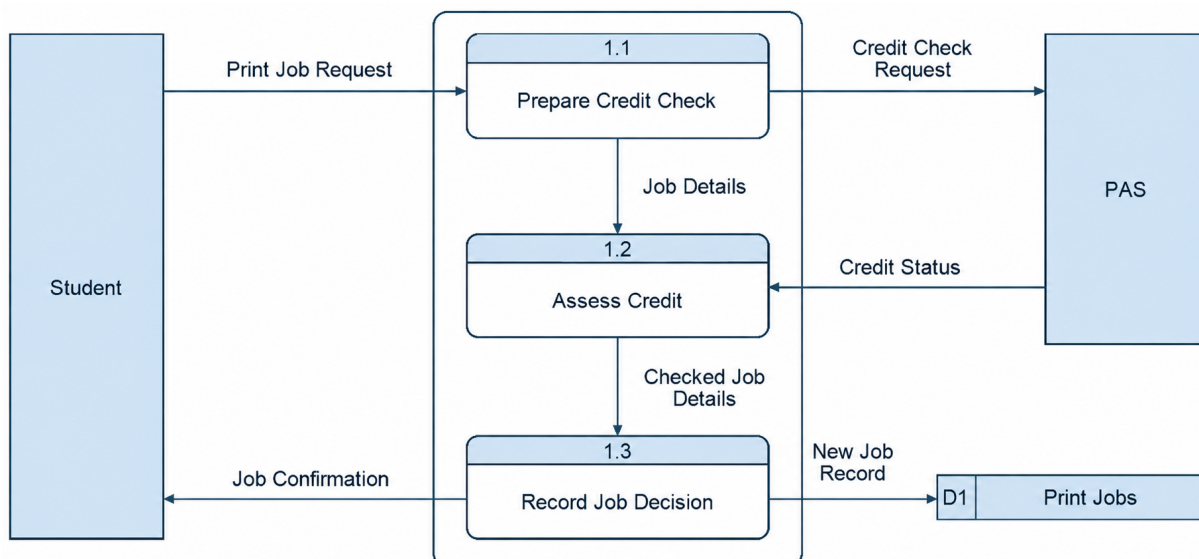
A.3 Draw your Level 1 DFD

Connect each context flow to the process responsible for it, then add store reads and writes. Each status update carries the job ID and new status.



A.4 Decompose a process into Level 2

Open process 1 into three subprocesses. Keep its five boundary flows and add the internal flows needed to pass the request and credit result.



Source	Data flow	Destination	Contents
1.1	Job Details	1.2	Document, student ID and printing options
1.2	Checked Job Details	1.3	Request data and credit result

Job Confirmation reports acceptance or rejection. Only accepted jobs produce a queued New Job Record; rejected requests are not stored. The outer outline groups subprocesses.

A.5 Check balancing and review the diagram

Balancing means preserving the parent's inputs and outputs in the more detailed diagram. These rows show how to record each match; check the remaining flows together.

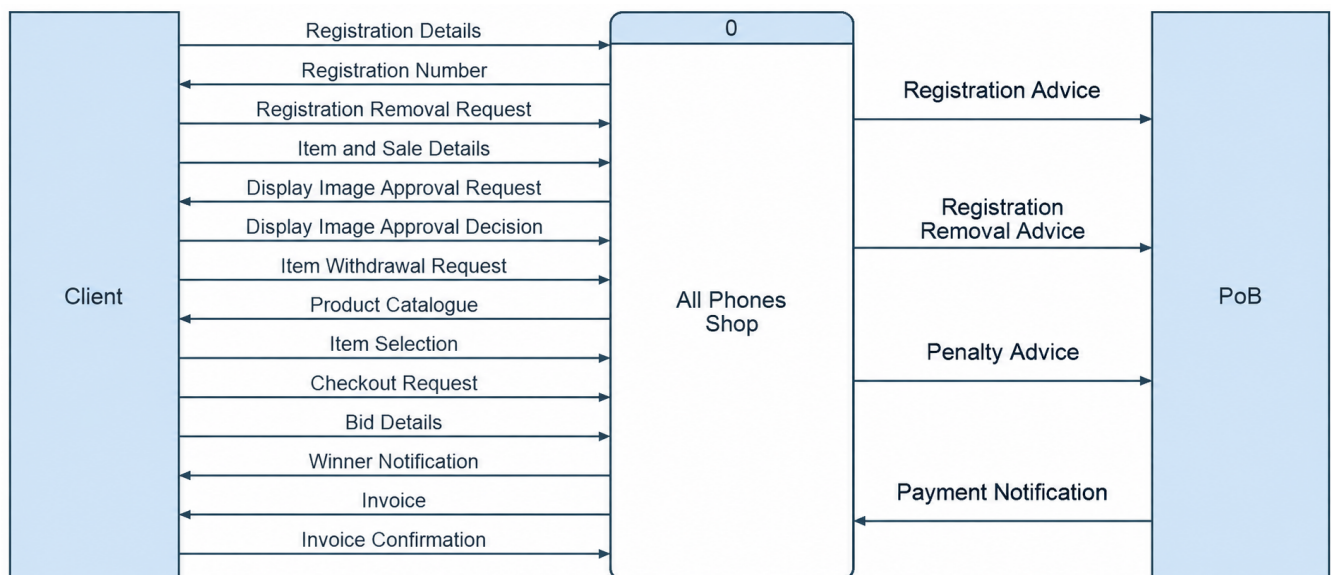
Parent flow and direction	Child process and direction	Balancing result
Student → Print Job Request → 0	Student → Print Job Request → 1	Matches
1 → New Job Record → D1	1.3 → New Job Record → D1	Matches

Check all seven context flows and five parent-process flows. The flow from 1.1 to 1.2 is internal to this decomposition. Review example: if New Job Record is missing, add the flow from 1.3 to D1.

Part B All Phones Shop Level 1 DFD

B.1 Review the supplied context diagram (10 minutes)

Part B is assessed individually. Use the supplied Level 0 diagram and the APS case from Lab 3. Compare the diagram with your earlier work and record any differences. Develop your own processes, stores, internal flows and Level 2 decomposition.



Client includes individual and corporate clients. Only corporate clients request registration removal. Penalty Advice covers withdrawal and overdue payment. PoB handles payments and financial accounts; login is outside scope.

B.2 Identify processes and stores (20 minutes)

Use the case description to decide how APS should be divided into processes and what data it needs to store. Give each process a name and ID, explain its responsibility, and describe the contents of each data store and which processes read or update it. Support your design choices with evidence from the case.

B.3 Draw your Level 1 DFD (40 minutes)

Draw your Level 1 DFD in Visual Paradigm using your own process and store design. Preserve the supplied context flows and add the internal data flows needed to meet the case requirements. Label every flow and use consistent IDs.

Build a context-to-Level-1 balancing table while drawing. Use the columns shown in A.5 and include every parent input and output. Record the business rules and any assumptions you make. Save APS Level 1 and export aps-level-1.png.

B.4 Decompose the auction process (40 minutes)

Choose the Level 1 process responsible for auction handling and decompose it into Level 2. Use its actual ID as the prefix for your subprocess IDs. Identify and name the subprocesses yourself, explain their responsibilities, and show the data they exchange.

Preserve every input and output of the chosen parent process, including flows to stores and other processes. Keep shared store IDs. Explain how your decomposition meets the auction requirements in the case.

Create APS Auction Level 2 in Visual Paradigm. Build its balancing table while drawing. Save the project and export aps-auction-level-2.png.

Submission

Use the final 15 minutes to prepare and submit these four files under your-repo/04-dfd/.

File	Contents
lab04-solution.docx	Your name and student ID, the supplied context diagram and any recorded differences or corrections, process responsibilities, the data store table, both DFDs, both balancing tables, and business rules and assumptions.
aps-level-1.png	Complete APS Level 1 DFD.
aps-auction-level-2.png	Auction Level 2 DFD.
aps-dfd.vpp	Editable Visual Paradigm project with both diagrams.

Commit and push all four files, then check that they appear in your repository.

Assessment 10 points

Criterion	Points
Level 1 processes cover the case and have clear responsibilities	2
Stores and internal flows supply the data each process needs	2
Context and Level 1 balance, supported by the comparison table	1.5
Auction Level 2 has suitable subprocesses and balances with its parent process	2
Correct notation, clear names and directions, readable diagrams	1
Business rules and assumptions are recorded	0.5
Complete submission in the required formats	1